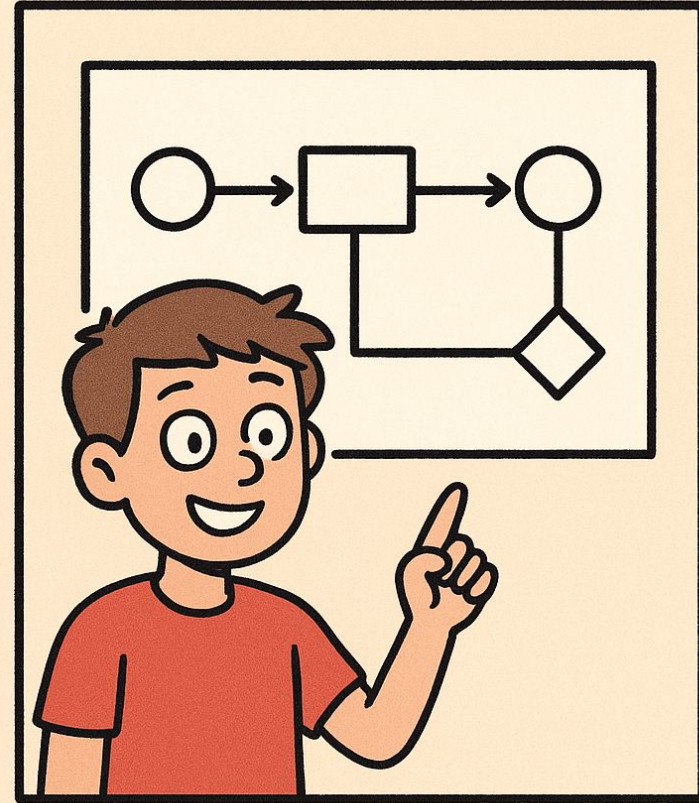
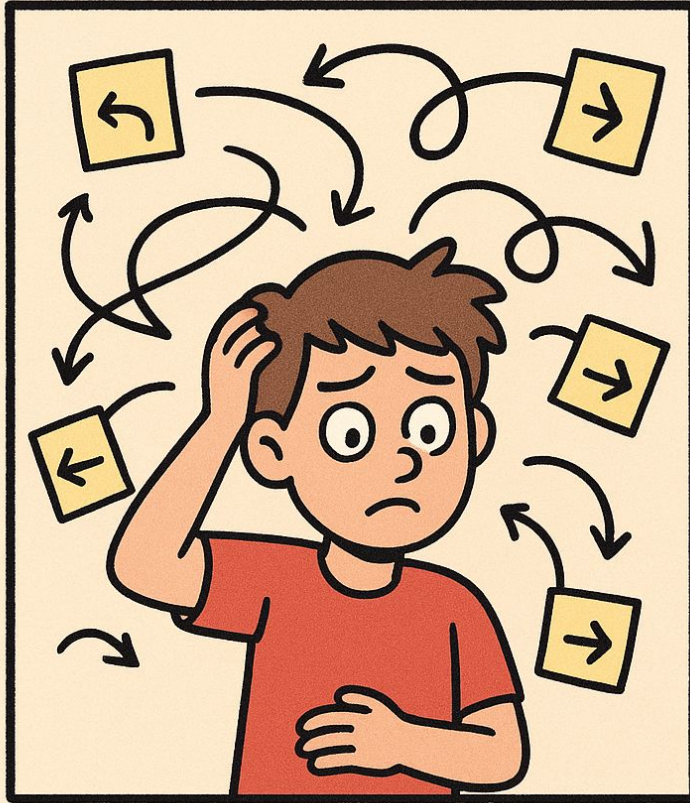




FROM CHAOS TO CLARITY WITH BPMN!

Announcements



NEWS

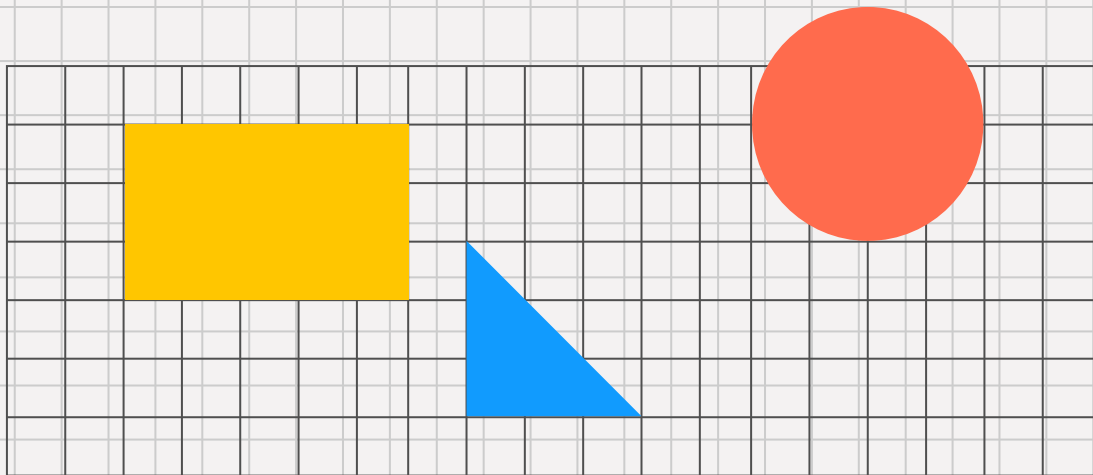
- Project teams are now final
- Task #1 presentations
 - 2/23 - Teams 1, 2, 3
 - 2/25 - Teams 4, 5, 6
 - 3/2 - Teams 7, 8, 9
 - 3/4 - Teams 10, 11
- Assignment #1 due 2/9
- **Quiz #1 on 2/11 - on Lectures 3&4**

Spring 2026

IS 365

Lecture 4:

Advanced Modeling
Concepts



Task #1: Presentation Outline



1. **Introduce the business** - location, product / services offered, annual revenue, growth, key challenges
2. **Describe the business process** - video clips or text summaries of interviews, other online research on the business process that you plan to fix
3. **Show the business process architecture** of the current workflow and clearly highlight inefficiencies
4. **Provide an estimate** of the dollar impact (on revenue, profitability, etc.) of correcting these inefficiencies through automation
5. **Show detailed business process models** - “as-is” process, “to be” process

Take Survey 2

The screenshot displays a course management interface with a grid background. It features three main sections: 'Lecture Slides', 'Project', and 'Assignments'. Each section has a header bar with a green checkmark, a dropdown arrow, a plus sign, and a three-dot menu icon. The 'Assignments' section is expanded, showing a list of three items. The first item is 'Assignment 1: Process Identification' with a rocket icon, dated Feb 9, and worth 5 pts. The second item is 'Survey: IS365 Knowledge Adventure' with a rocket icon, dated Jan 21, and worth 1 pt. The third item, 'Survey 2: Process Identification', is highlighted with a red rectangular box; it also has a rocket icon, is dated Feb 4, and is worth 1 pt. Each item in the list has its own set of icons (three dots, rocket, green checkmark, and three dots) on the right side.

Section	Item	Icon	Date	Points	Status
Lecture Slides					✓
Project					✓
Assignments	Assignment 1: Process Identification	Rocket	Feb 9	5 pts	✓
	Survey: IS365 Knowledge Adventure	Rocket	Jan 21	1 pts	✓
	Survey 2: Process Identification	Rocket	Feb 4	1 pts	✓

Survey 2: Answers

1. Which statement best describes Business Process Management (BPM)?

C. The art and science of overseeing how work is performed to ensure consistent outcomes and enable improvement

2. In the Ford Accounts Payable case, what was the primary driver for redesigning the process?

C. Lowering operational costs through workforce reduction

3. Why does BPM emphasize focusing only on *key processes* during process identification?

B. Monitoring and improving every process is too costly and resource-intensive

4. Which of the following is *not* one of the evaluation criteria used in the Process Identification phase?

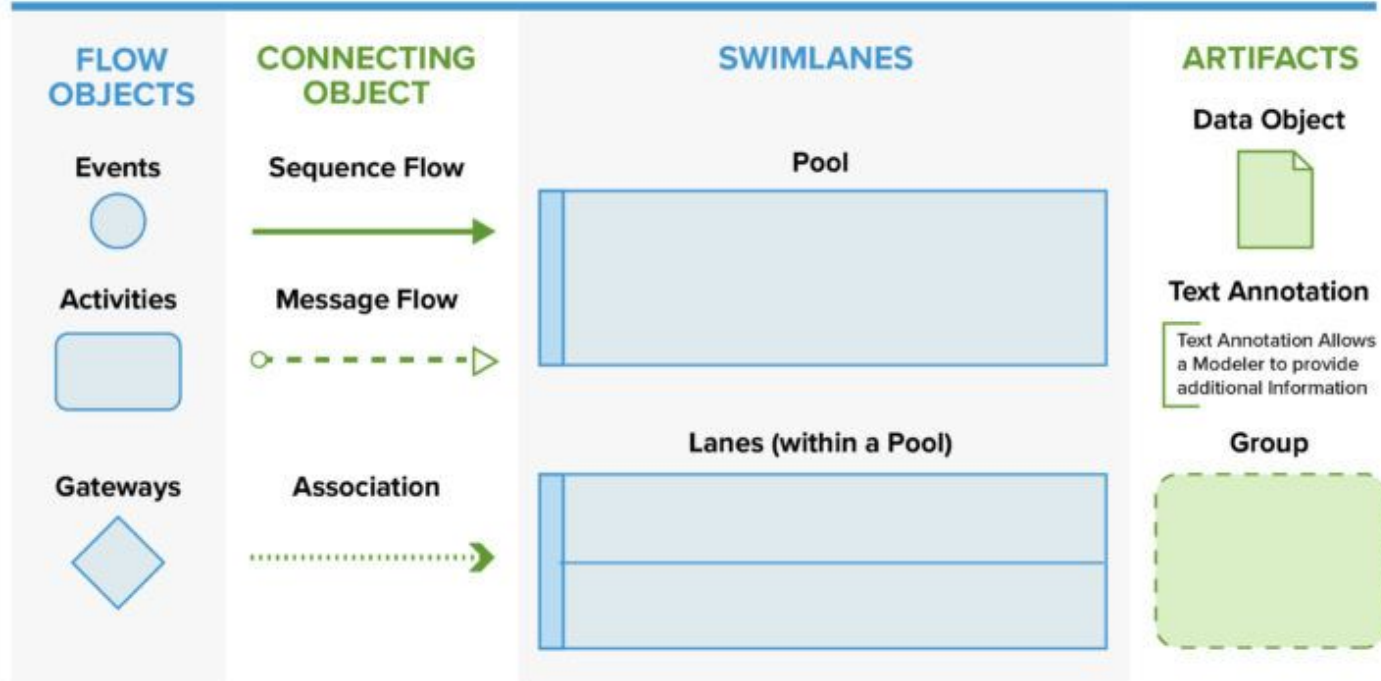
D. Choice of BPMN notation

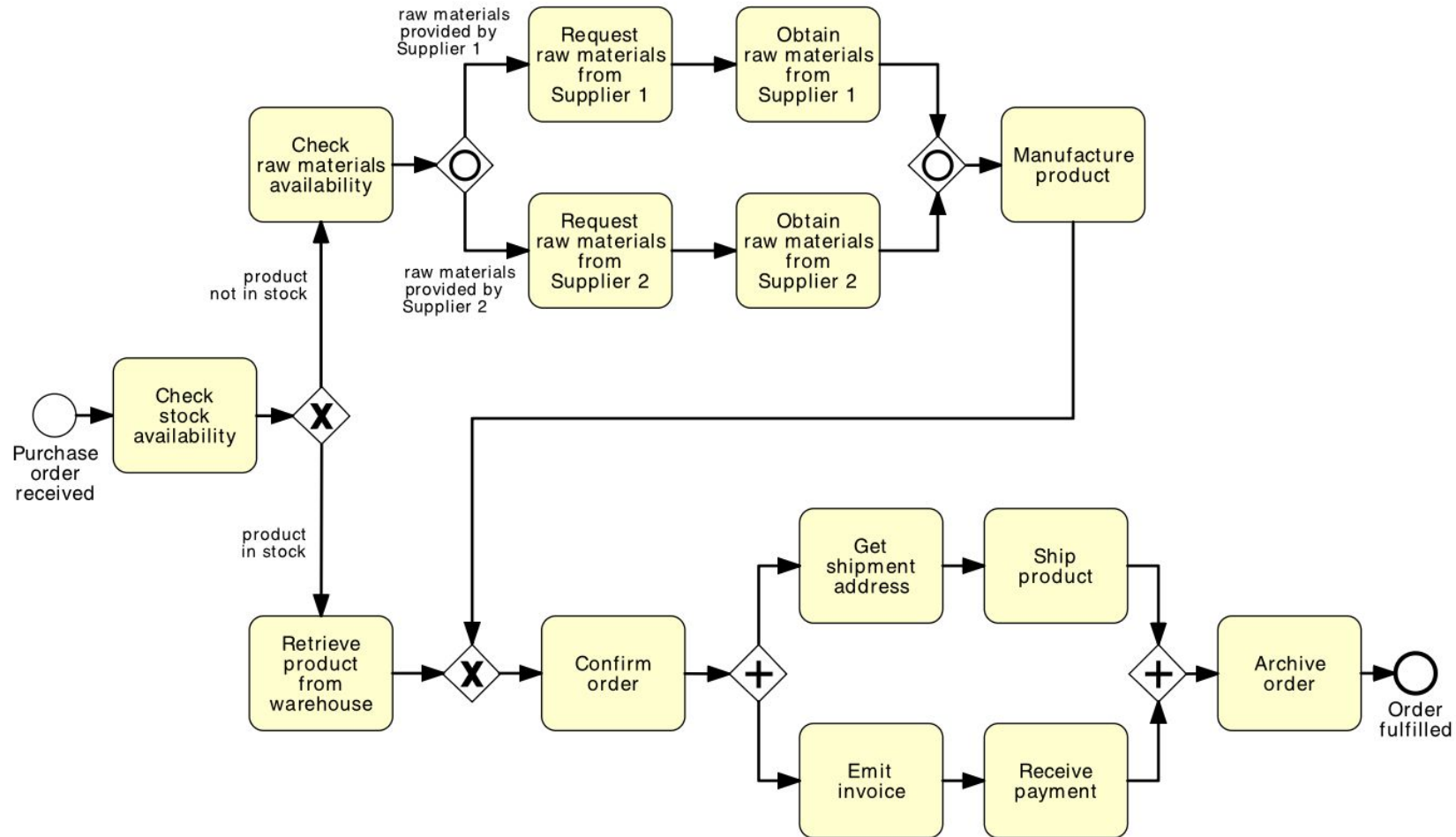
5. What is the main output of the Process Identification phase in the BPM lifecycle?

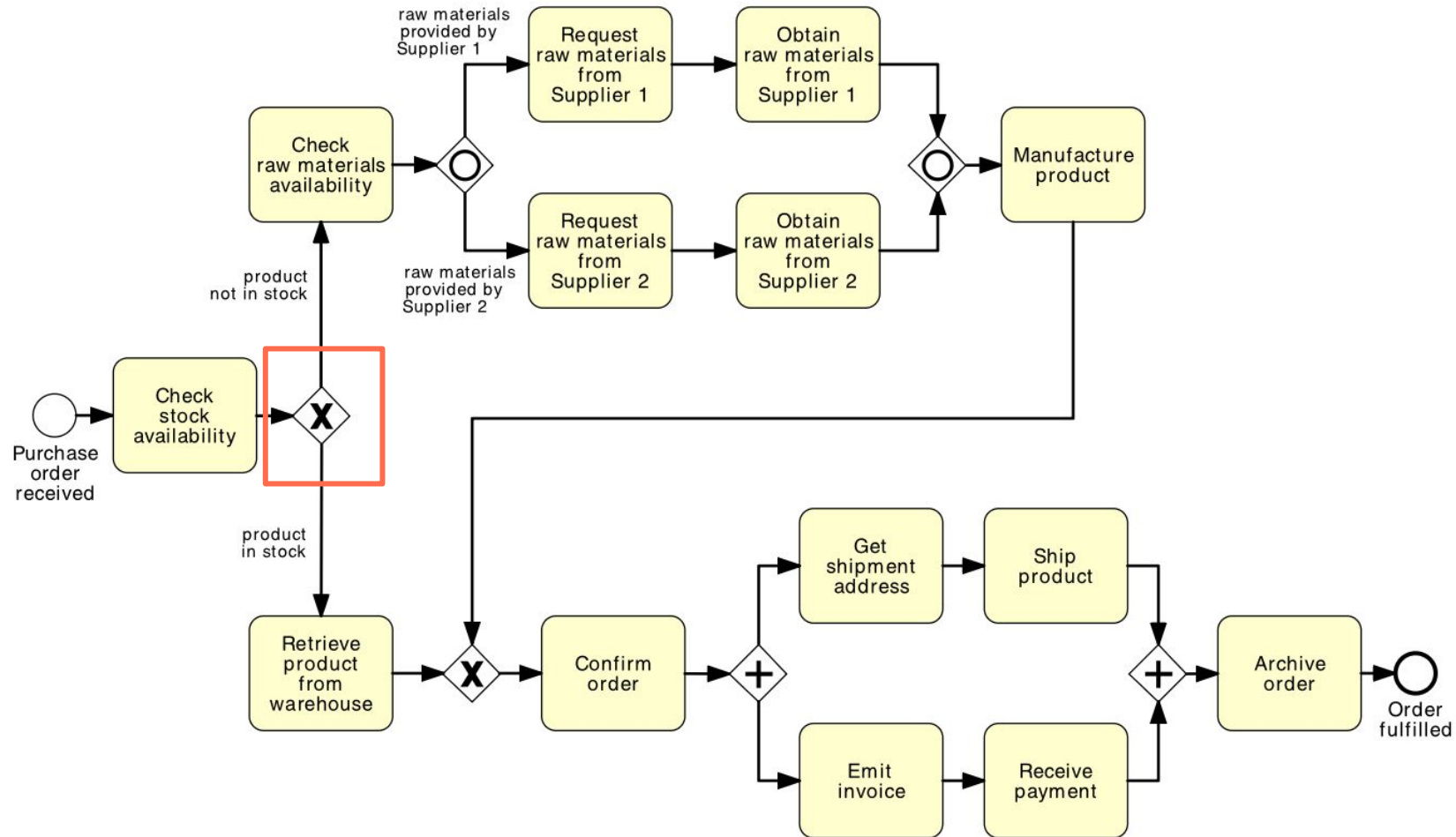
C. A prioritized process architecture

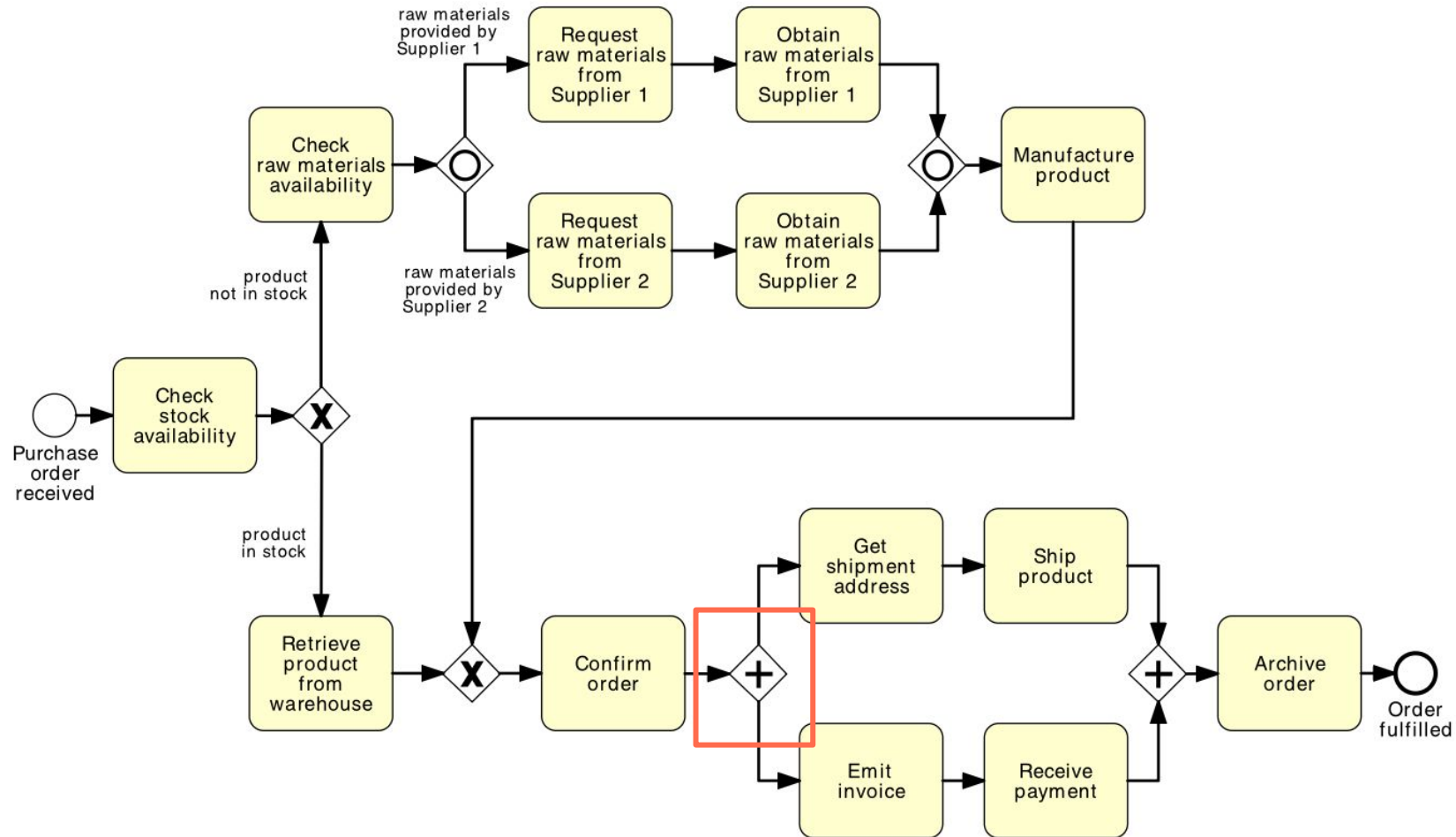
Last Class: Functional Perspective

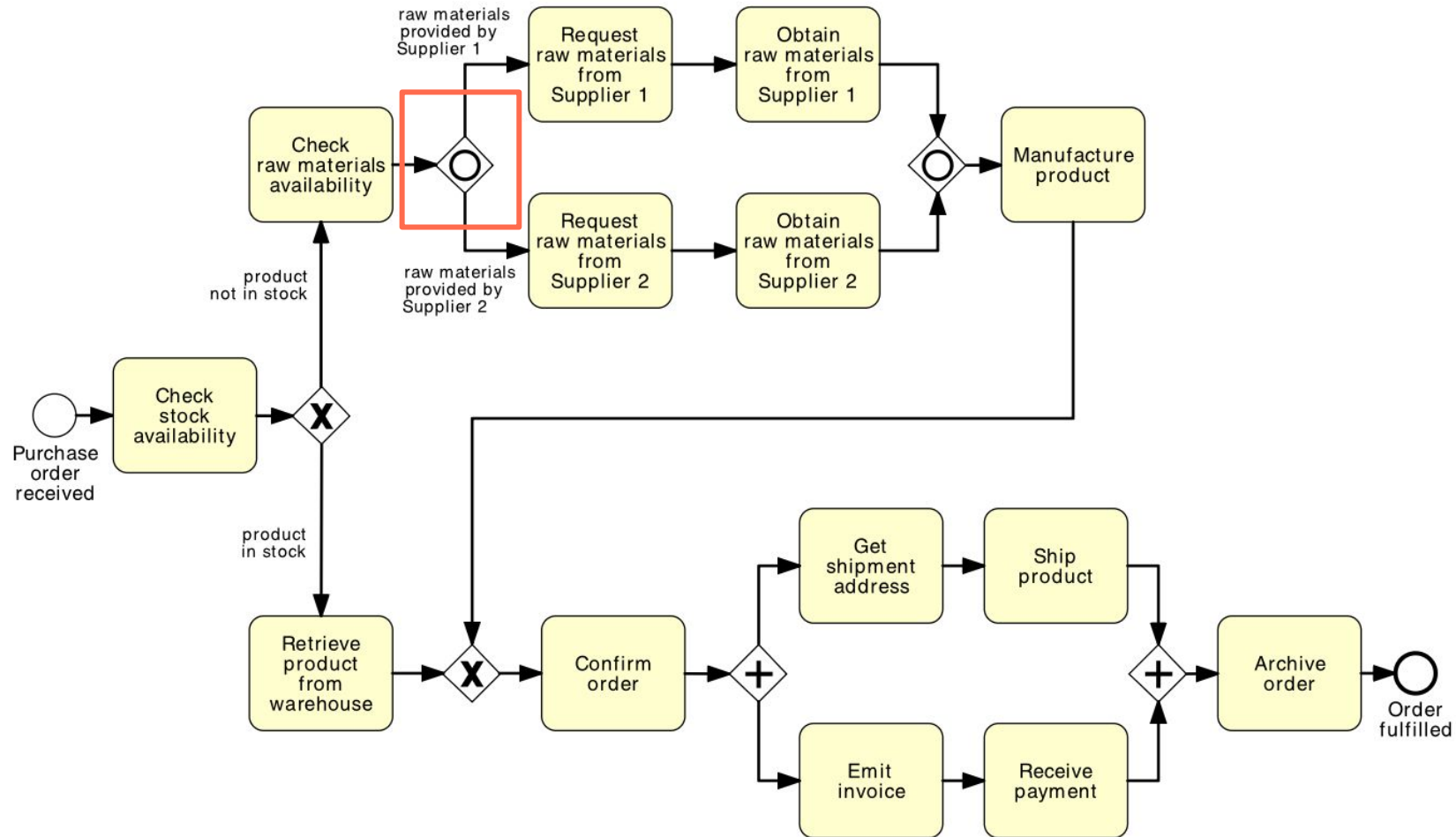
Core Set of BPMN Elements











Data Perspective

Data Perspective in BPMN

- Captures **information flow** in processes
- **Data objects:** documents, emails, files, forms
- Represented with the document symbol

Data Objects and Associations

- **Inputs:** required before an activity executes
- **Outputs:** produced by an activity
- **Associations:** dotted arrows (do not replace sequence flows)



Message Flow



Data Association



Data Input



Data Output



Data Object



Data Store



Subprocess

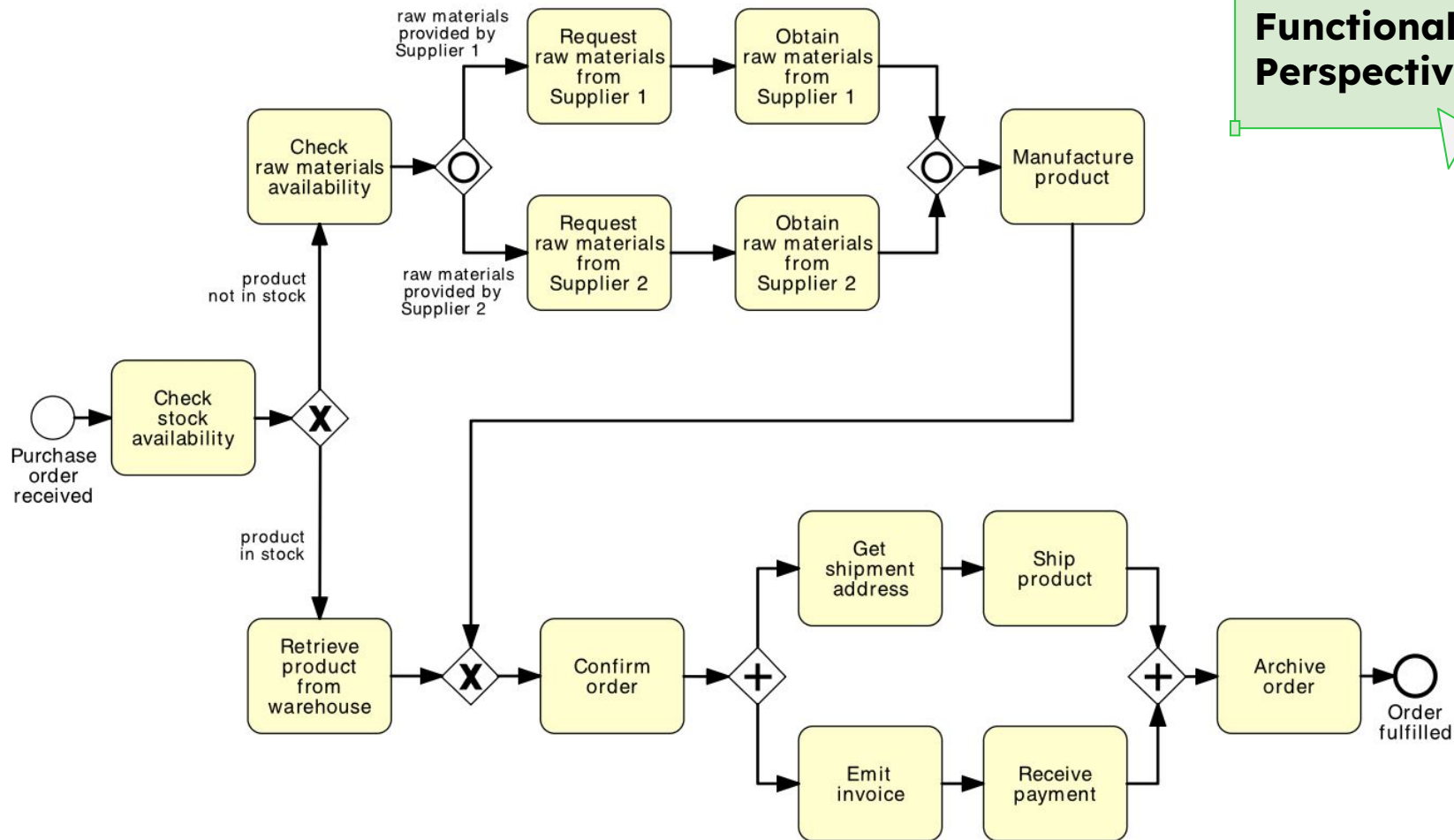
States and Data Stores

- Data objects can **change states** (e.g., "Order [confirmed]")
- Data **stores are persistent** (e.g., databases, filing cabinets)
- Activities can read/write from stores

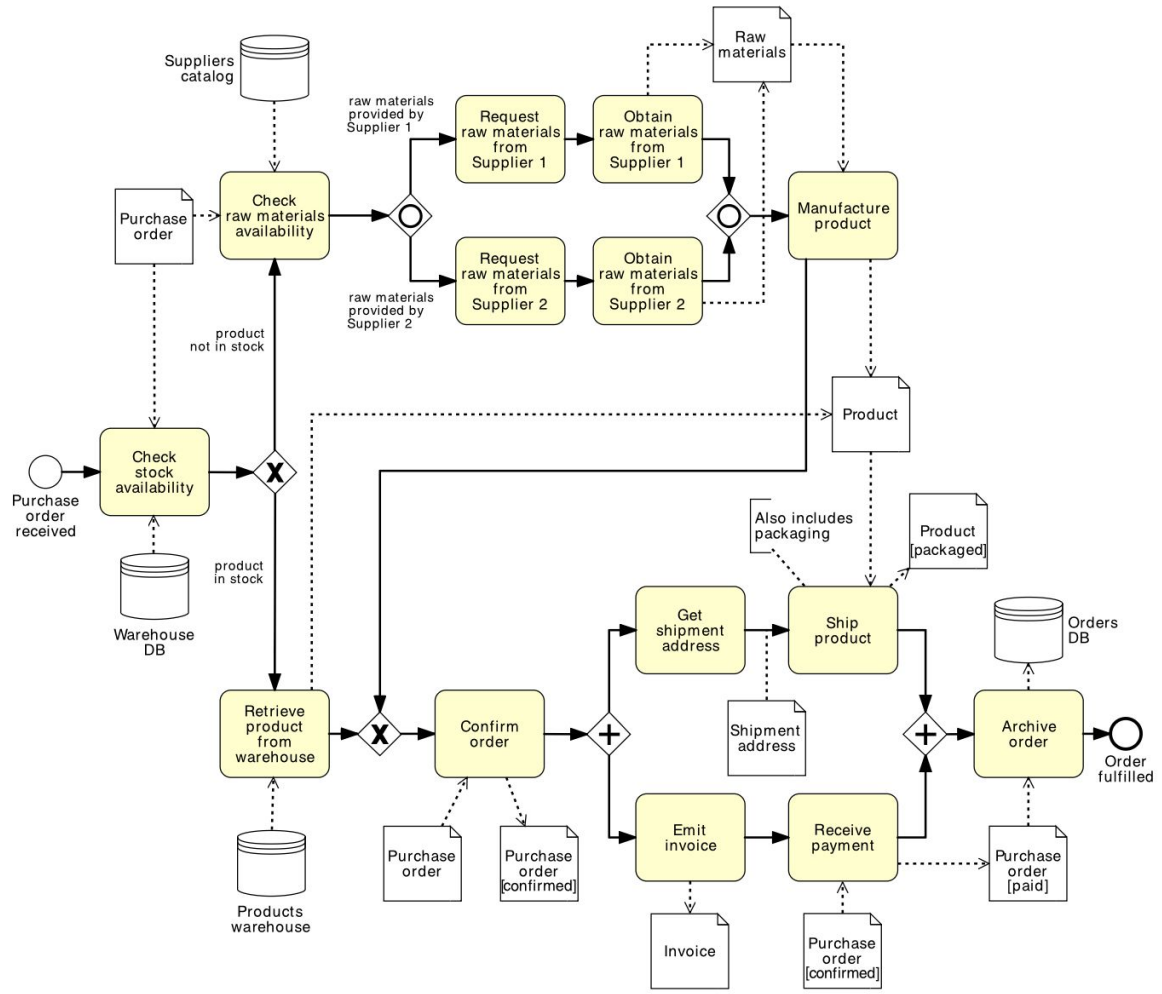
Annotations

- Provide clarifying information or business rules
- Improve readability, no effect on execution

Functional Perspective

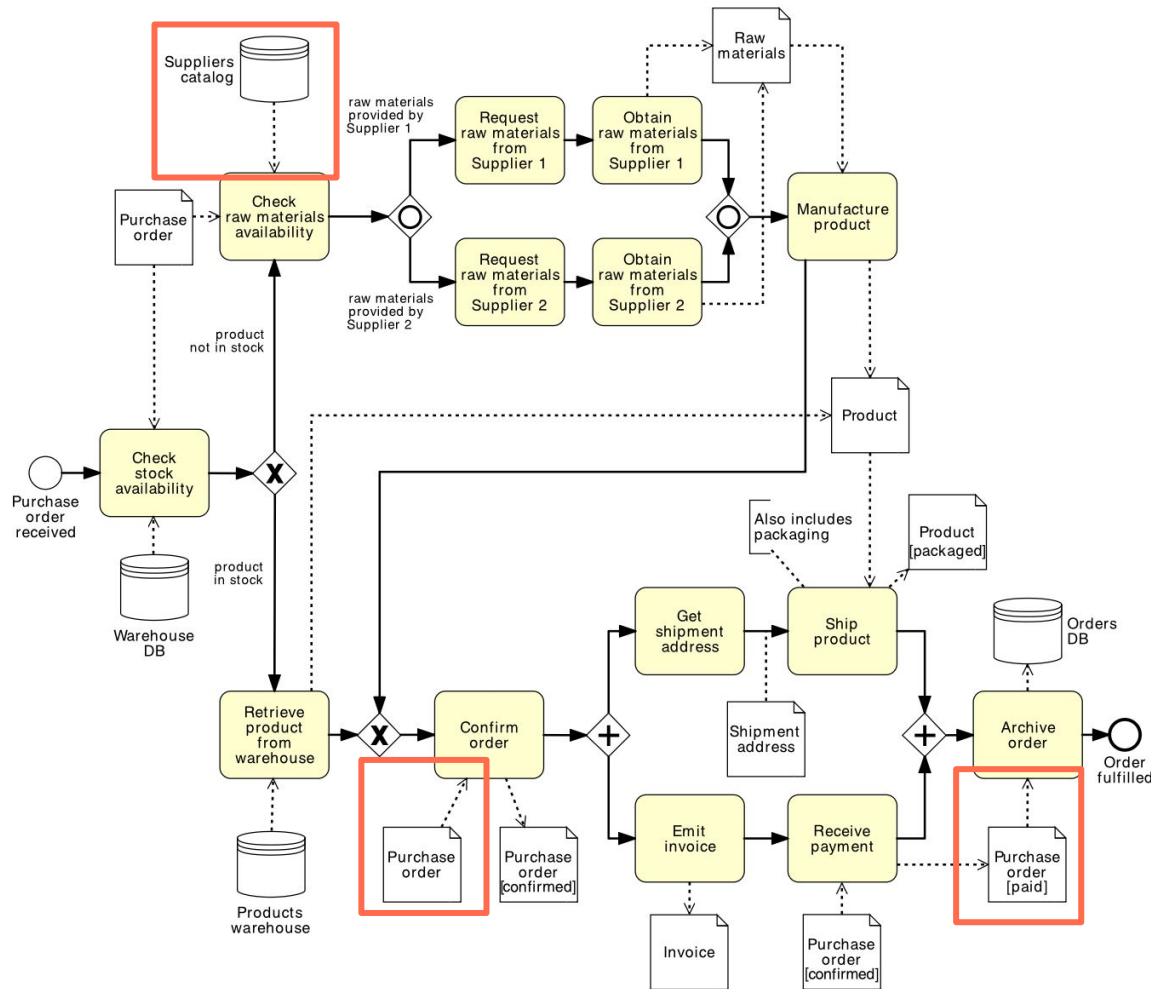


Data Perspective



Can you find?

- Data object
- Data store
- Message flow



Data Perspective

Can you find?

- Data object
- Data store
- Message flow

Resource Perspective

Active vs. Passive Resources in Processes

- **Active:** humans, systems, machines
(perform tasks)
- **Passive:** tools/equipment
(used but not performing tasks)

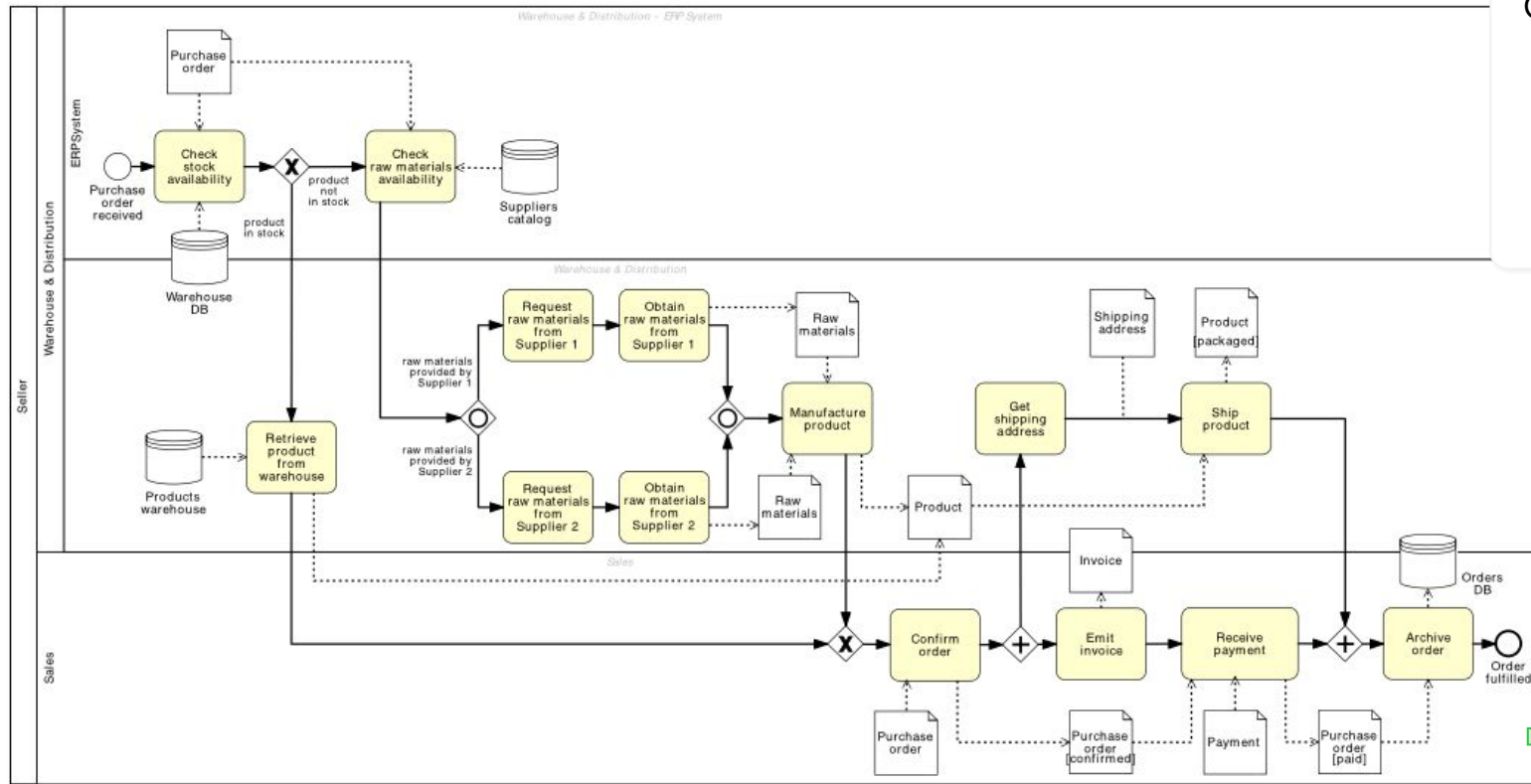
Pools and Lanes

- **Pools:** major participants (e.g., organizations)
- **Lanes:** departments, roles, or systems inside pools
- Help assign responsibilities



Can you find?

- Active resource
- Passive resource

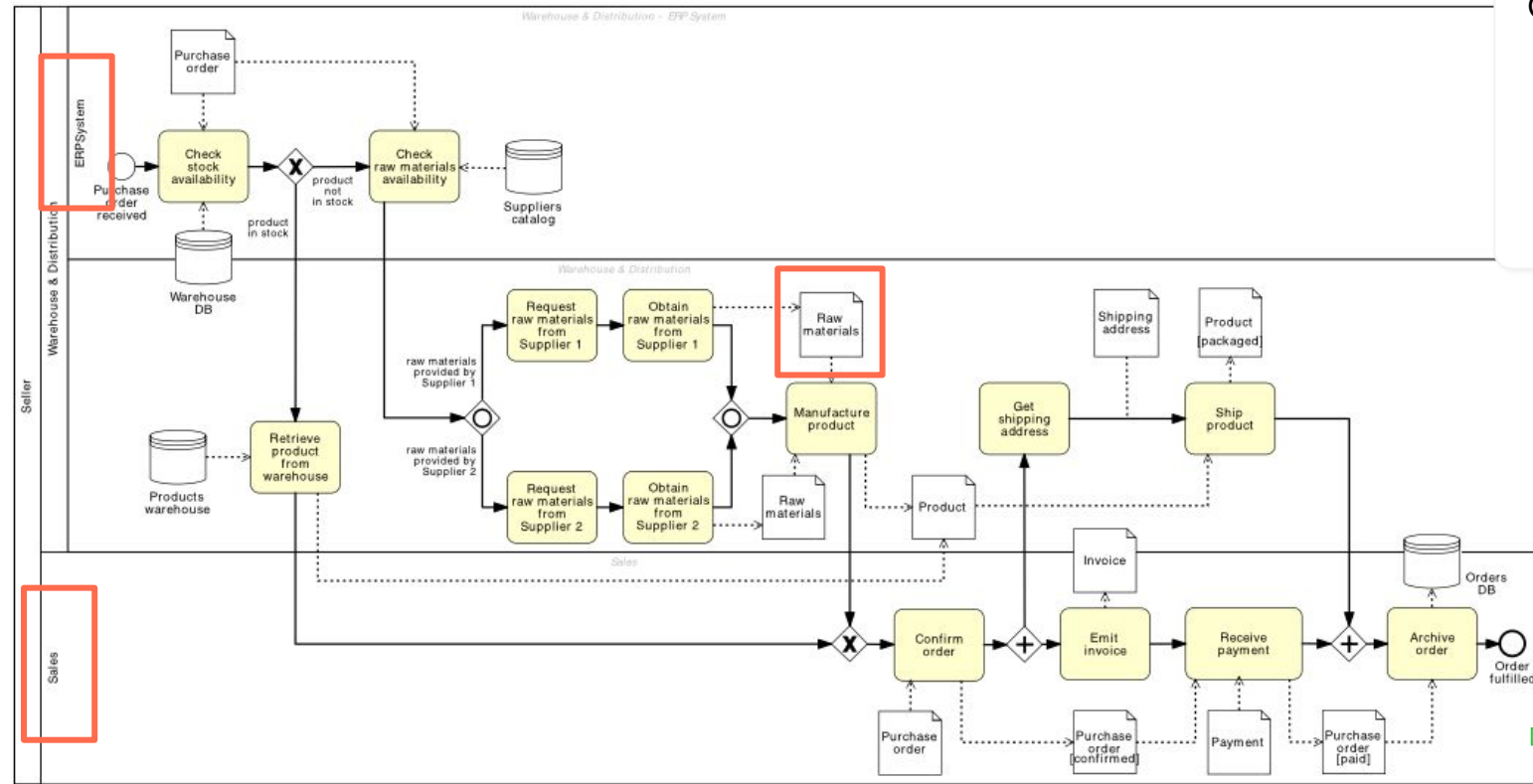


Resource Perspective



Can you find?

- Active resource
- Passive resource



Resource Perspective

Collaboration Diagrams

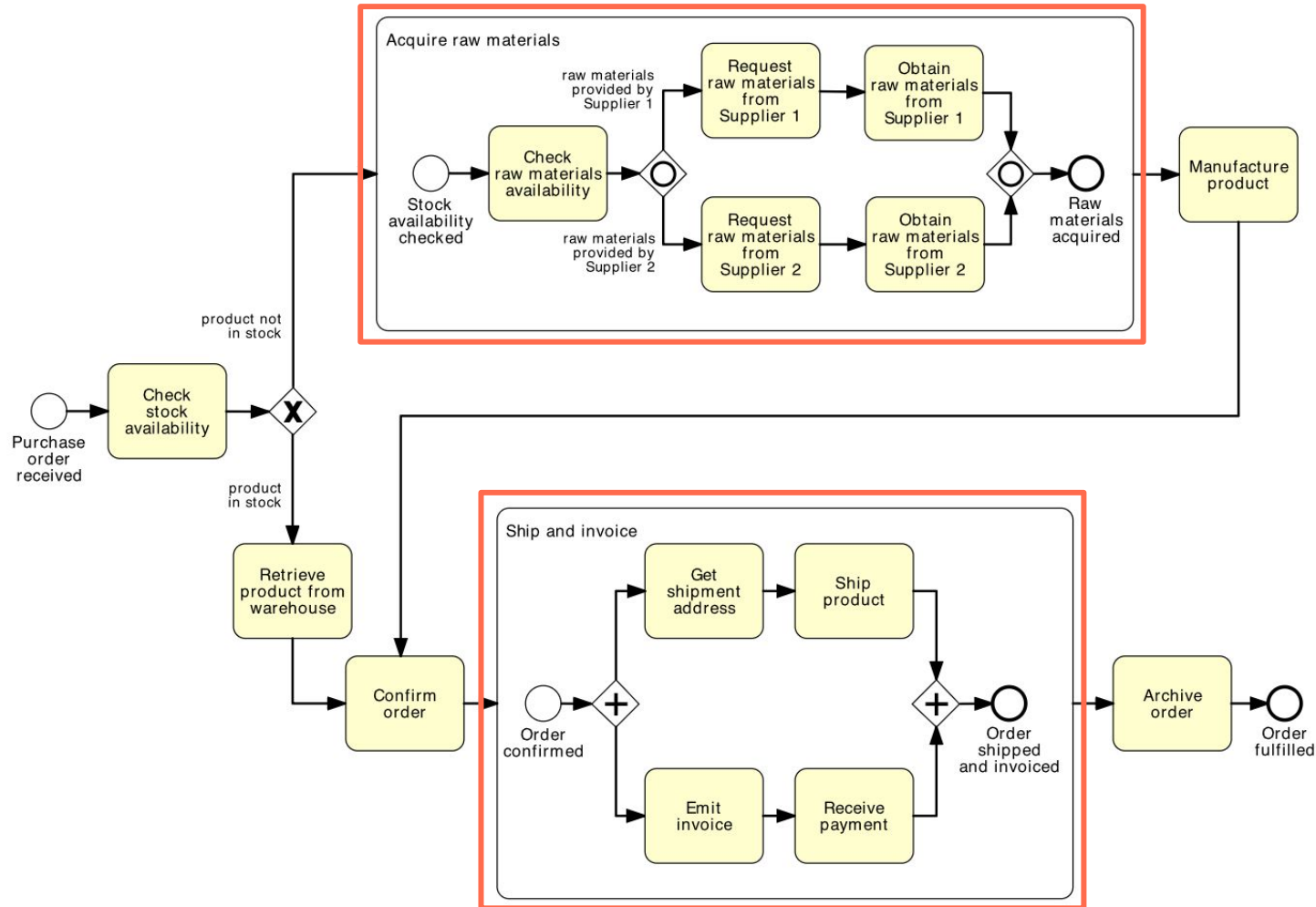
- Show interactions across organizations
- **Message flows:** communication between pools
- **White box pool:** internal details visible
- **Black box pool:** internal details hidden

Process Decomposition & Reuse

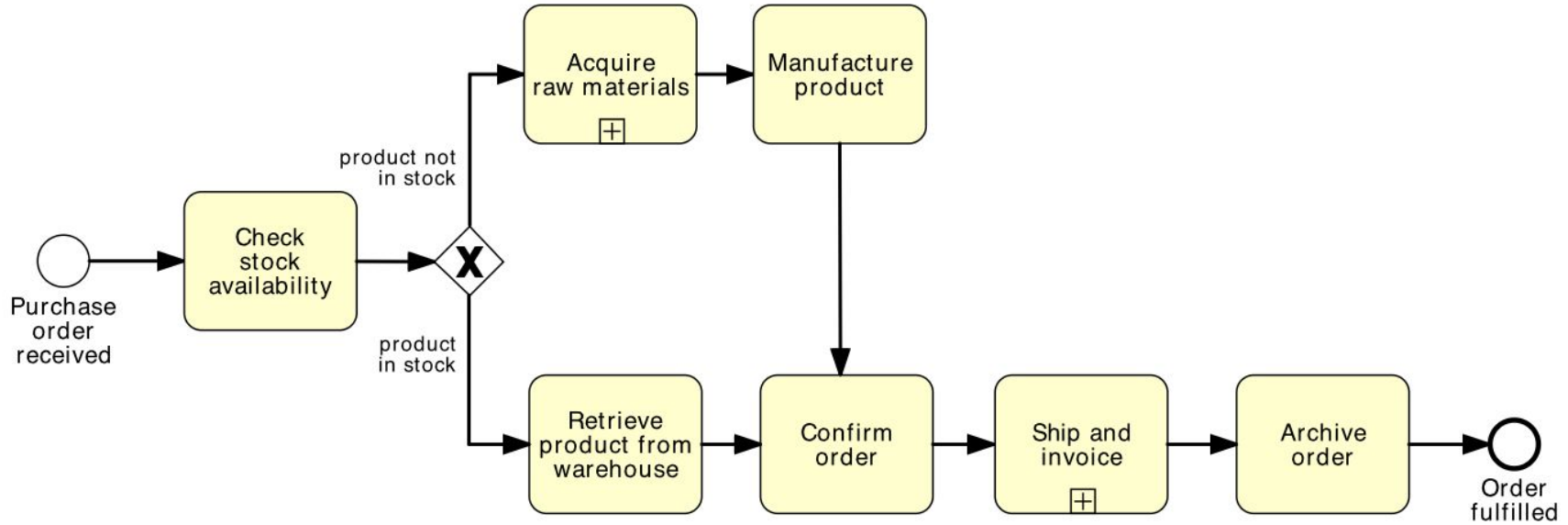
Process Decomposition

- Break large processes into smaller sub-processes for clarity and to focus on specific activities
- **Hierarchical decomposition:** add details progressively from core activities
- Reduce errors and improve overall model integrity

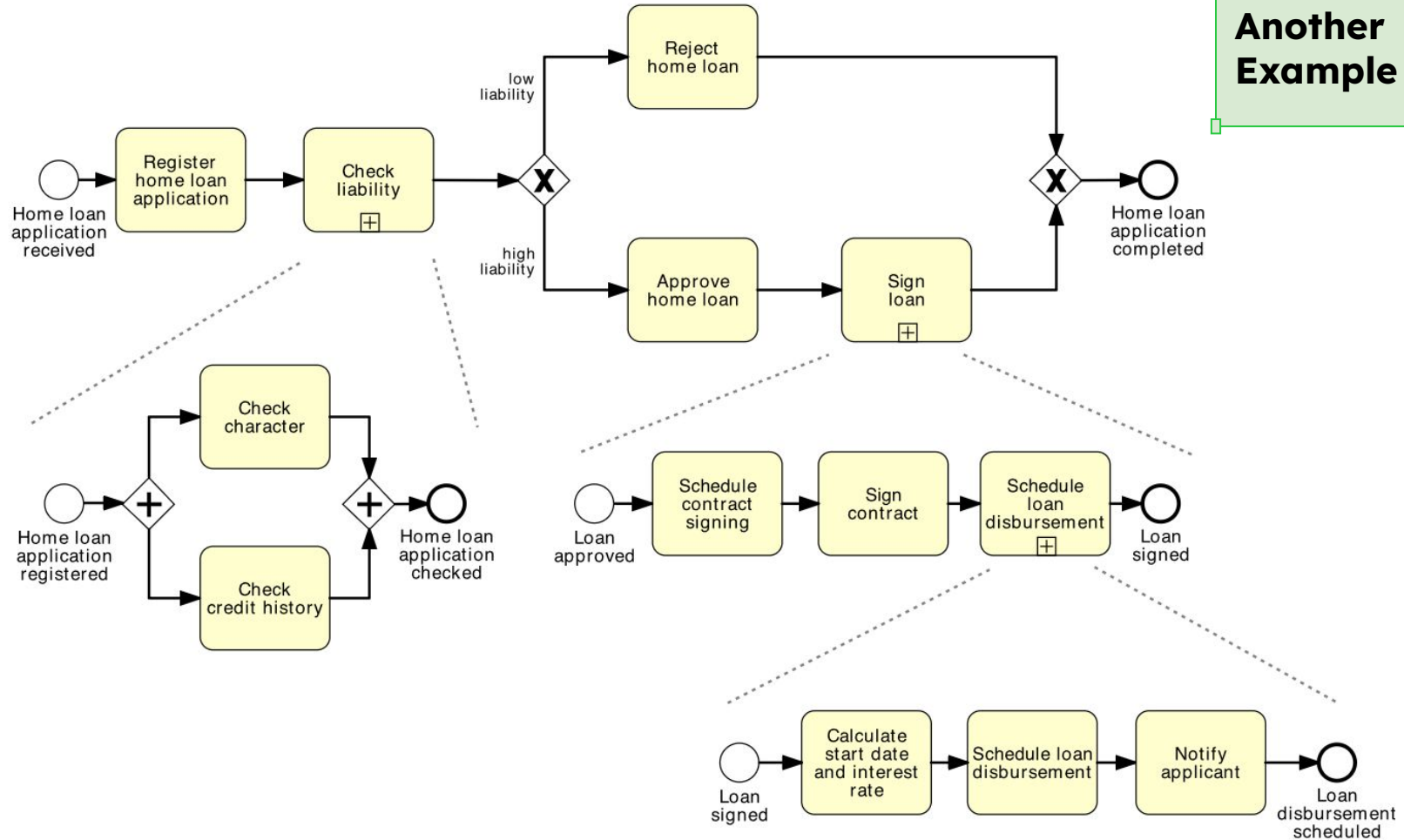
Before



After

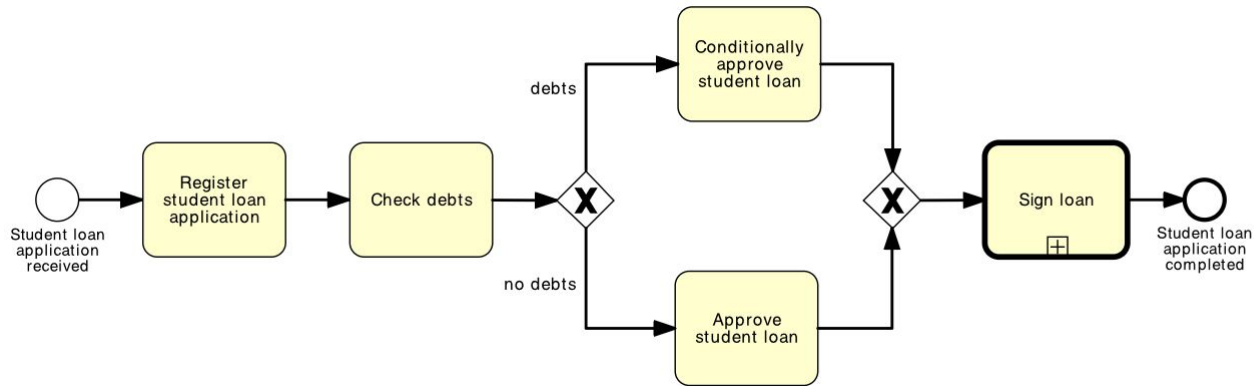
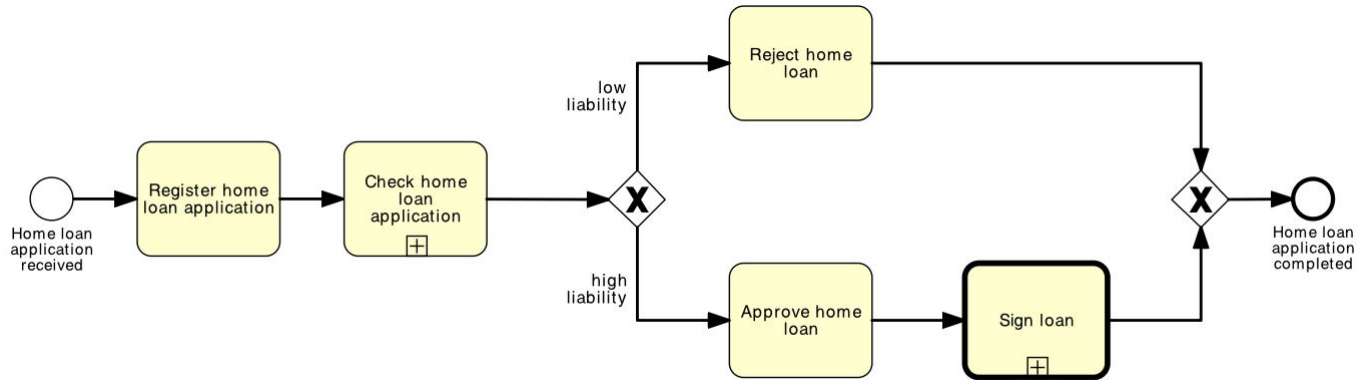


Another Example



Process Reuse

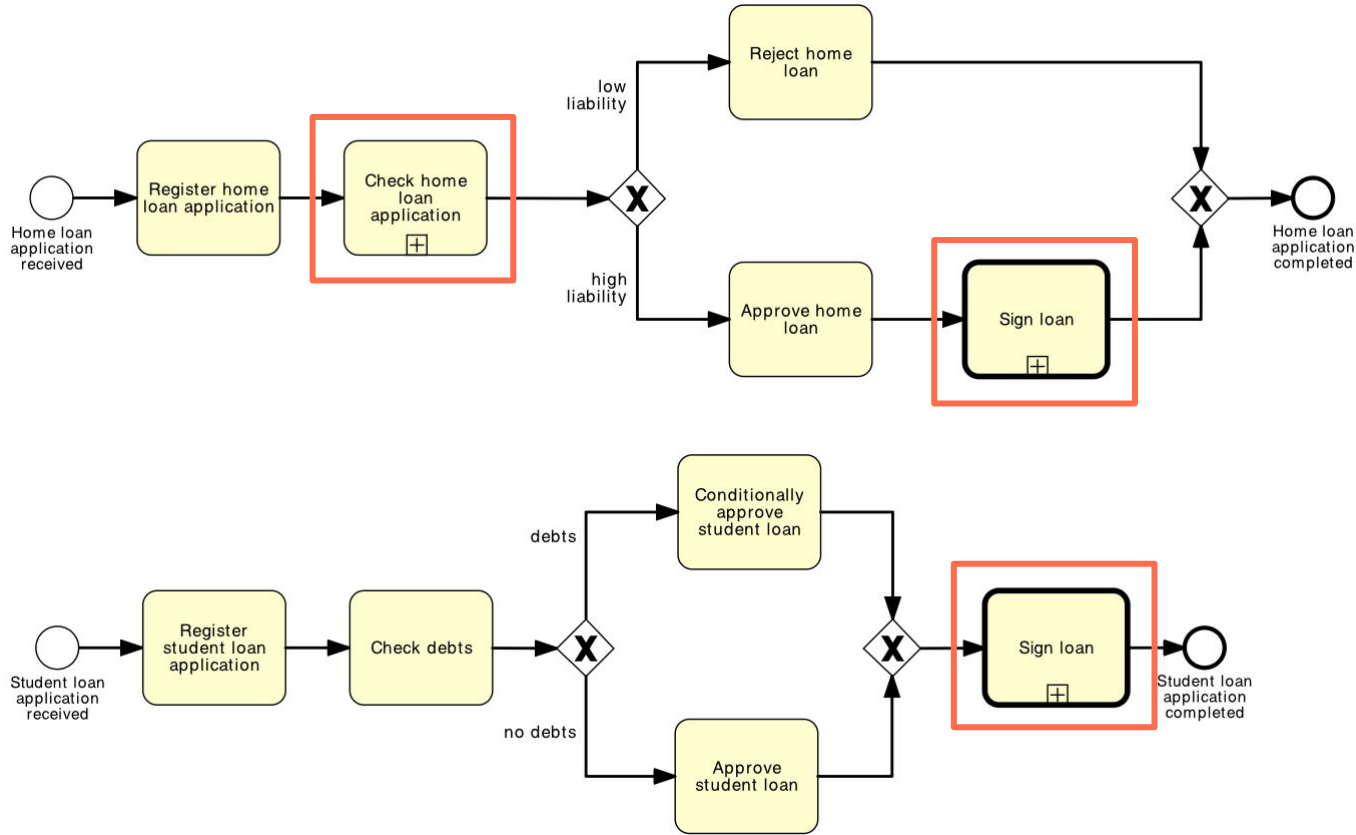
- Global sub-processes are independently defined and reused across multiple process models
- Promotes modularity and consistency
- Visually represented with thicker borders and a plus symbol



Can you find?

- Decomposed sub-process
- Reusable sub-process

**Decomposition
/ Reuse**



Can you find?

- Decomposed sub-process
- Reusable sub-process

**Decomposition
/ Reuse**

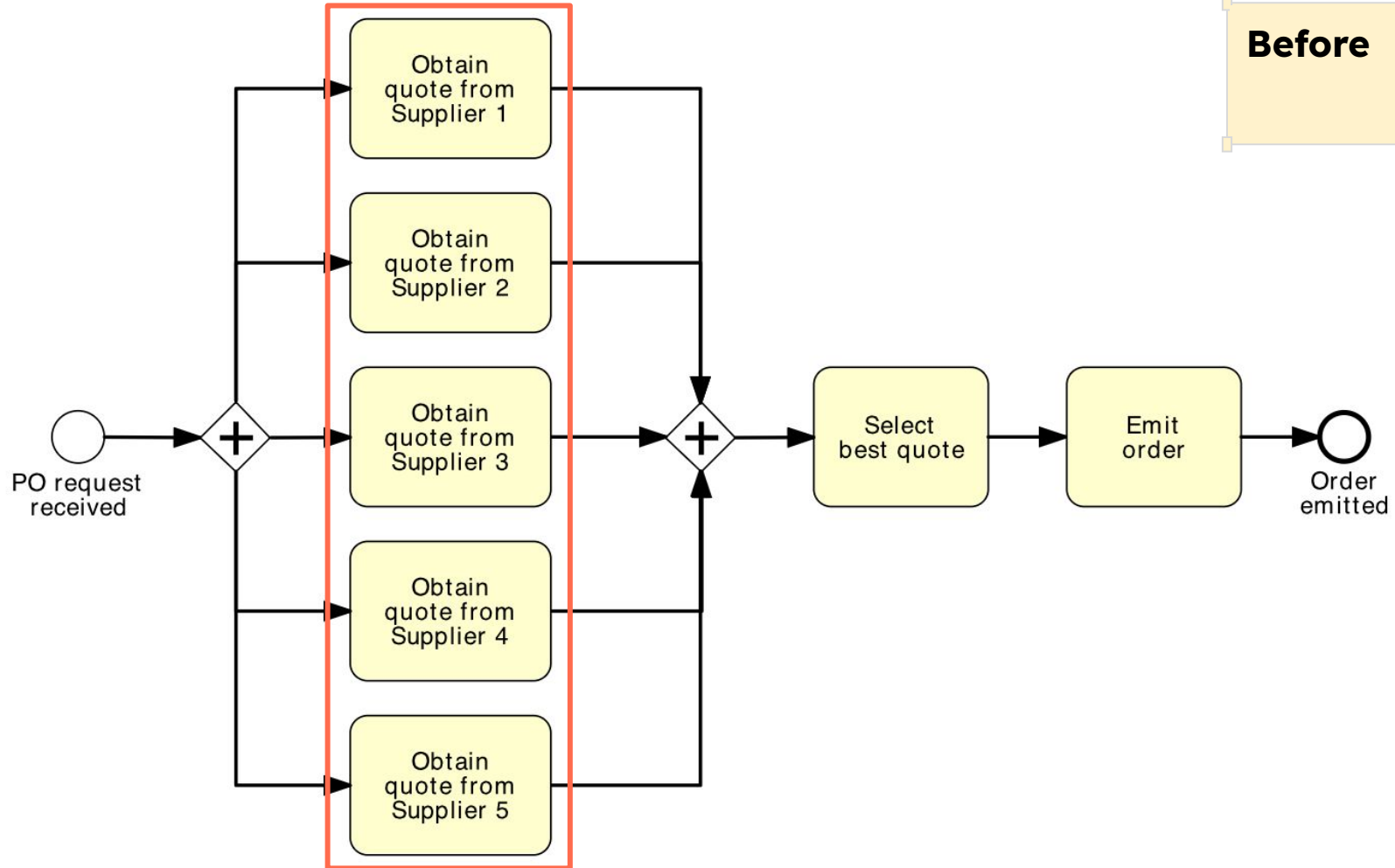
Rework & Repetition

Loop and Multi-instance Activities

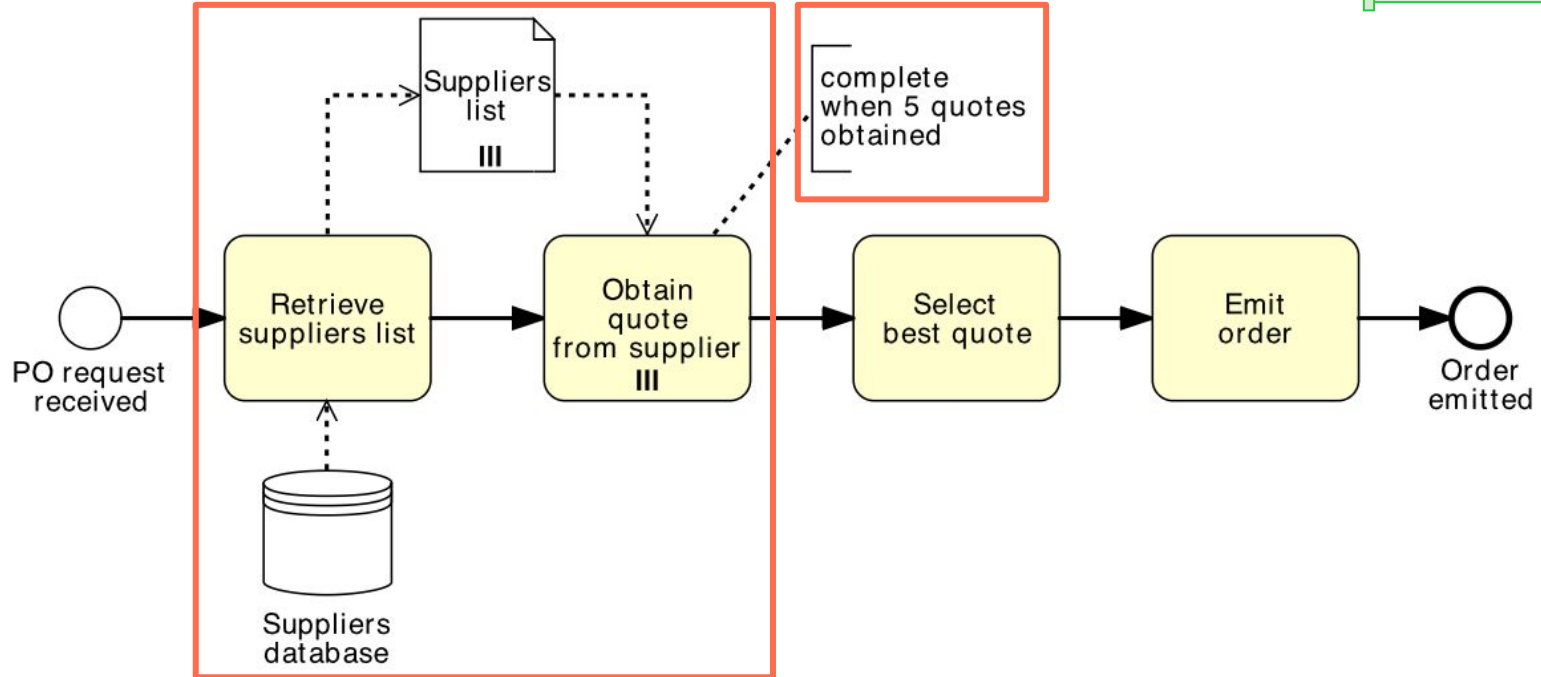


- **Loop activities:** model sequential task repetition until a specified condition is met → ideal for single entry and exit points
- **Multi-instance activities:** enable parallel execution of tasks across multiple entities, marked by three vertical lines
- **Multi-instance tasks:** dynamically adjust instance count based on input collections like supplier lists and completion conditions
- Reduce model size and enhance scalability

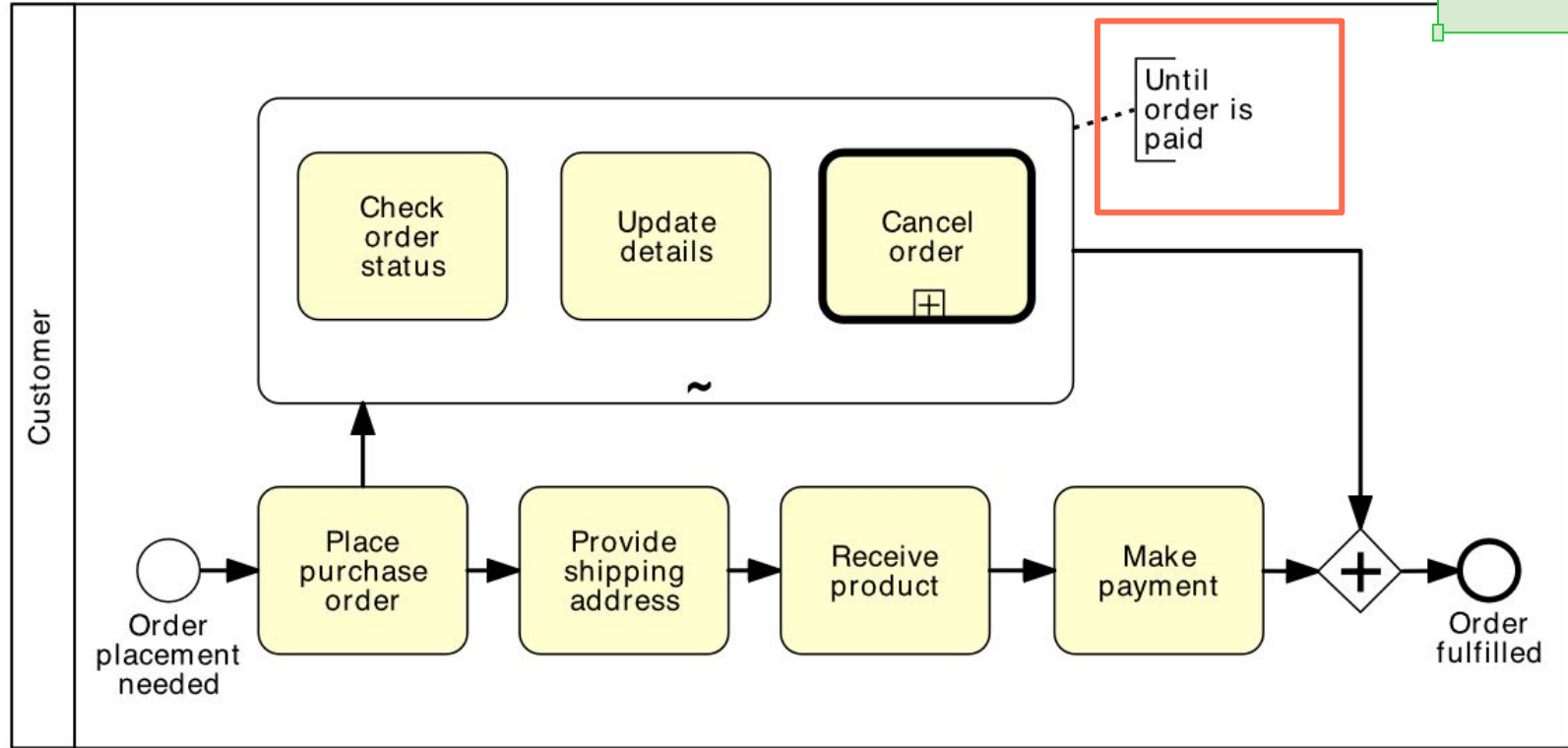
Before



After



Uncontrolled Repetition



Events & Exceptions

Types of Events

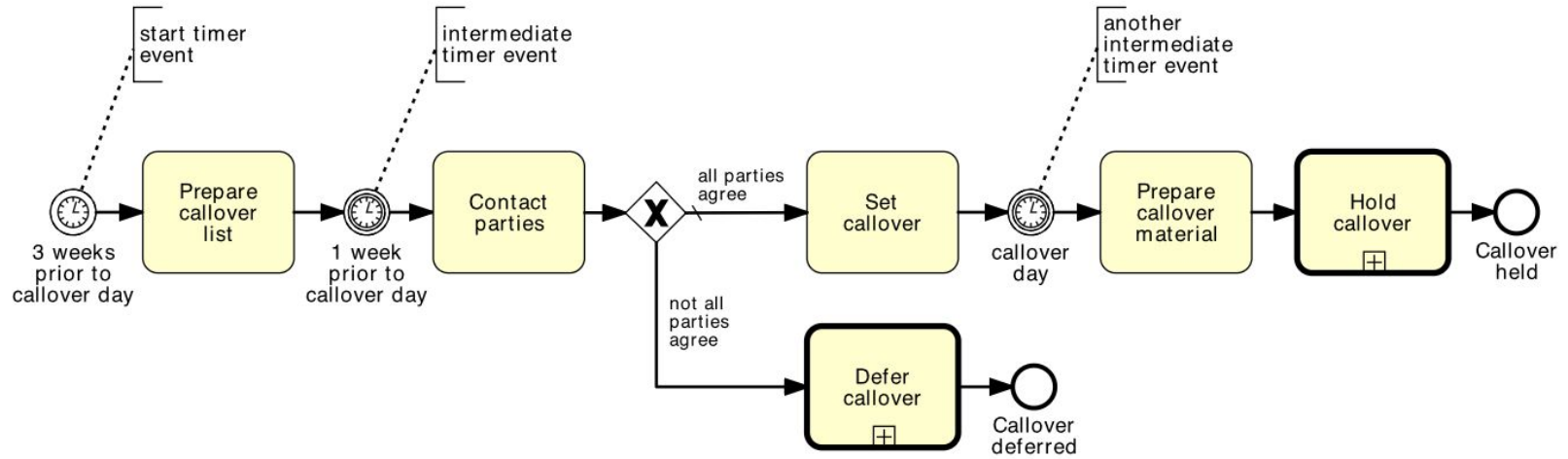
- Events can be start, intermediate, or end
- **Catch events:** wait for triggers and have unfilled symbols
- **Throw events:** emit triggers and are filled

Types of Events cont'd

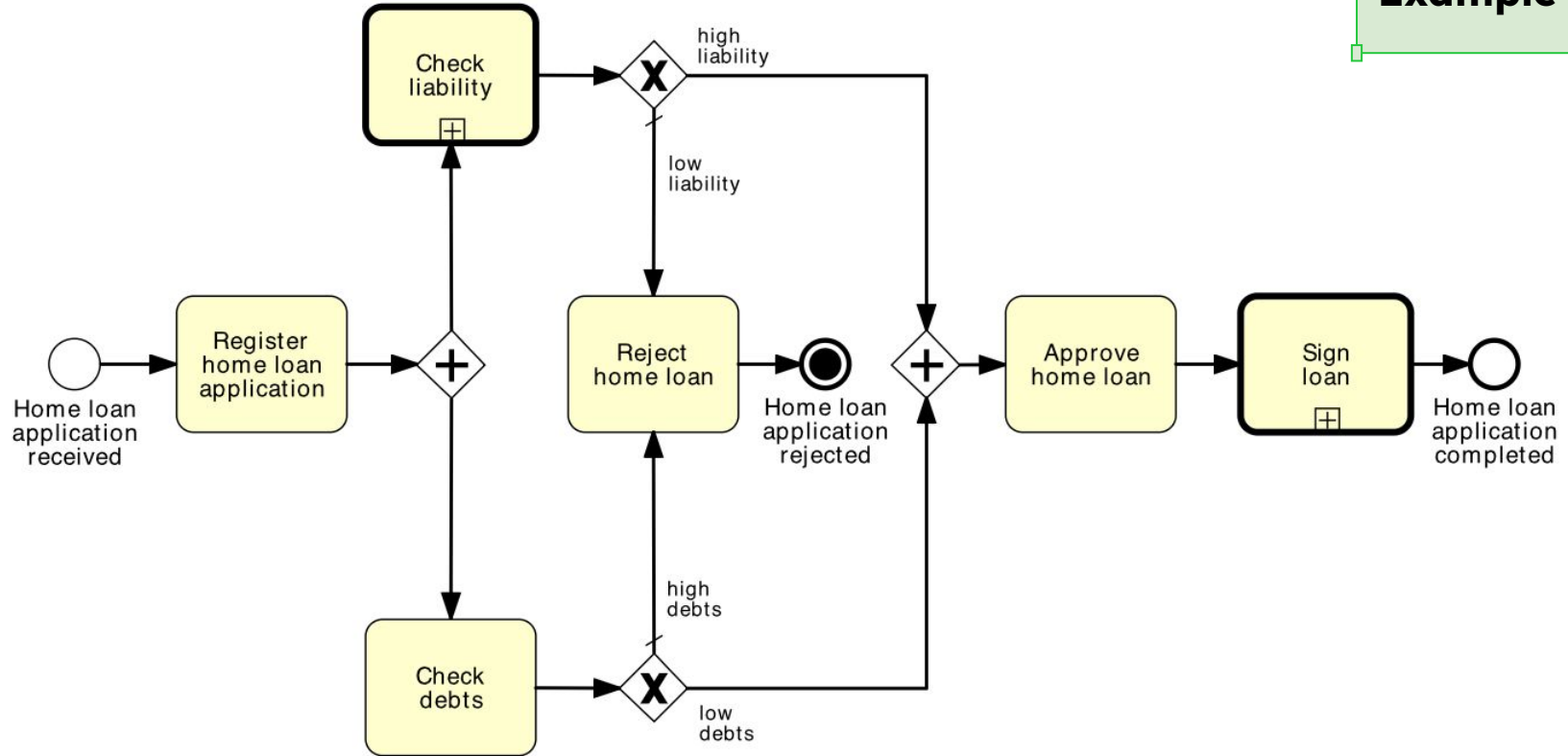
- **Message events:** enable communication between pools
- **Timer events:** handle temporal triggers like delays or schedules
- **Error events:** represent an error condition
- **Conditional events:** trigger flows based on business rules
- **Terminate events:** immediately terminate the process

Types	Start	Intermediate		End
	Top-Level	Catching	Throwing	
None				
Message				
Timer				
Error				
Conditional				
Terminate				

Example



Another Example



Choreographies

Choreography Diagrams

- Interactions as message exchanges between sender and receiver parties using bands and envelopes
- Choreography diagrams abstract from internal activities
- Focus on communication to ensure stakeholder alignment and interoperability
- Exclude data artifacts, enabling confidentiality and agreed interaction protocols

